Chapter 1 Review

Campbell Biology, AP Edition, 12e

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Summary of Key Concepts

Why is life considered to be an emergent property of a cell? **(BIG IDEA 4)**

Concept 1.1: The study of life reveals unifying themes

Organization Theme: New Properties Emerge at Successive Levels of Biological Organization

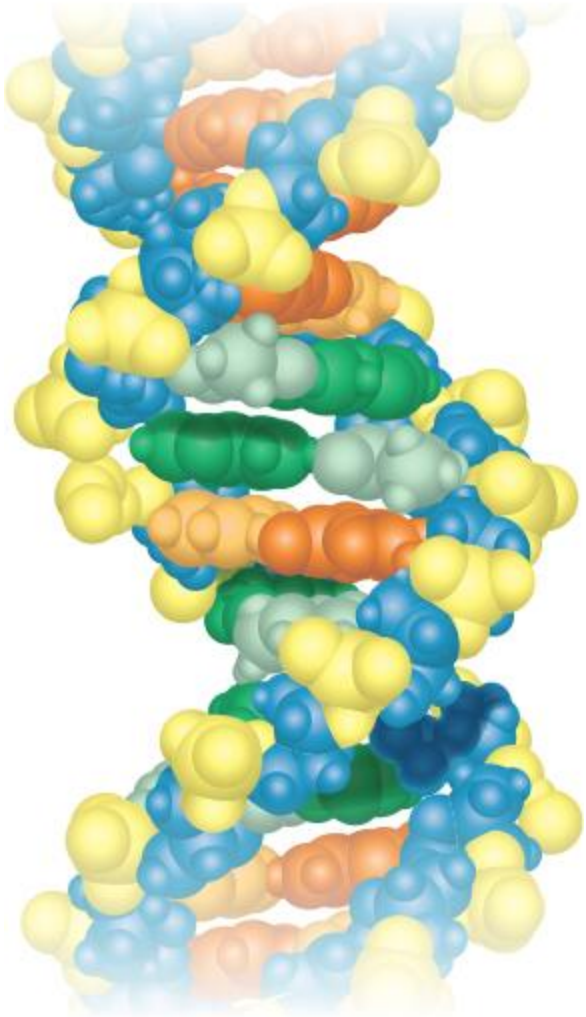
- The hierarchy of life unfolds as follows: biosphere > ecosystem > community > population > organism > organ system> organ > tissue > cell > organelle> molecule > atom. With each step from atoms to the biosphere, new **emergent properties** result from interactions among components at the lower levels. In an approach called reductionism, complex systems are broken down to simpler components that are more manageable to study. In **systems biology**, scientists attempt to model the dynamic behavior of whole biological systems by studying the interactions among the system's parts.

- Structure and function are correlated at all levels of biological organization. The cell, an organism's basic unit of structure and function, is the lowest level that can perform all activities required for life. Cells are either prokaryotic or eukaryotic. **Eukaryotic cells** have a DNA-containing nucleus and other membrane-enclosed organelles. **Prokaryotic cells** lack such organelles.

Information Theme: Life's Processes Involve the Expression and Transmission of Genetic Information

- Genetic information is encoded in the nucleotide sequences of **DNA**. It is DNA that transmits heritable information from parents to offspring. DNA sequences (called **genes**) program a cell's protein production by being transcribed into mRNA and then translated into specific proteins, a process called **gene expression**. Gene expression also produces RNAs that are not translated into protein but serve other important functions. **Genomics** is the large-scale analysis of the DNA

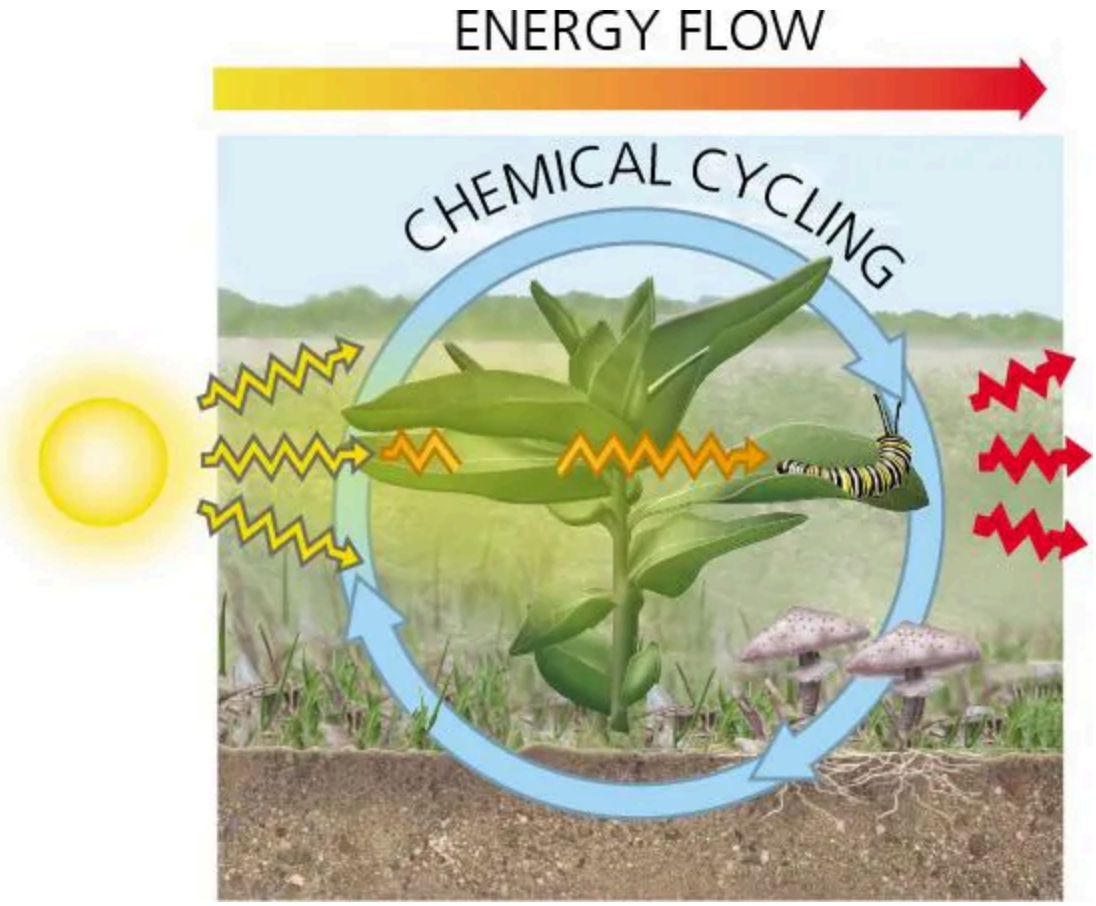
sequences of a species (its **genome**) as well as the comparison of genomes between species. **Bioinformatics** uses computational tools to deal with huge volumes of sequence data.



Energy and Matter Theme: Life Requires the Transfer and Transformation of Energy and Matter

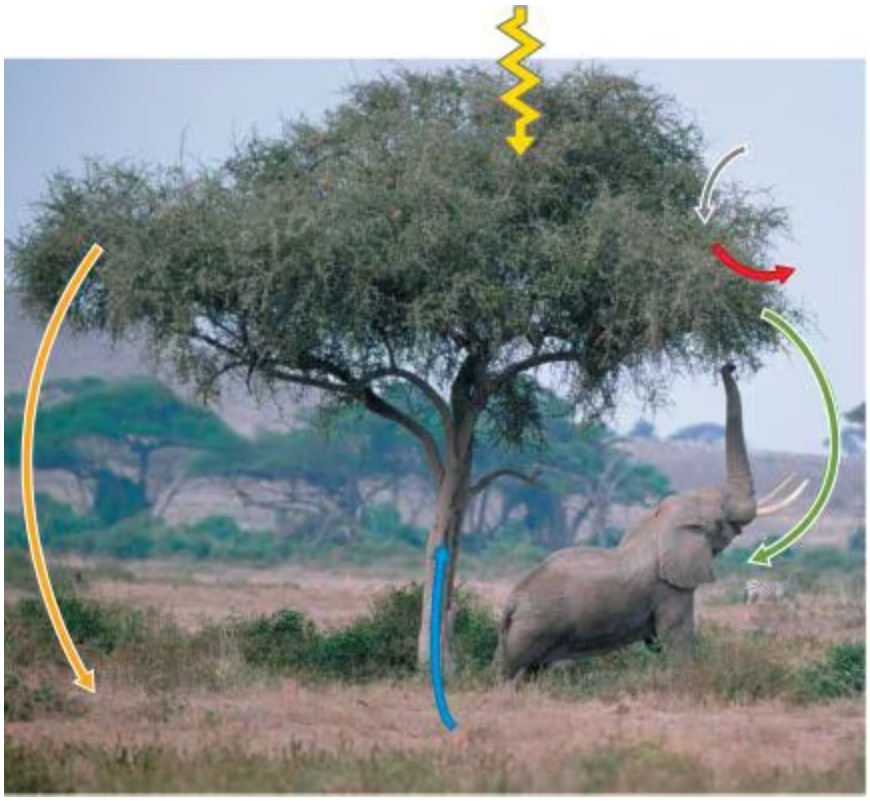
- Energy flows through an ecosystem. All organisms must perform work, which requires energy. **Producers**convert energy from sunlight to chemical energy, some of which is used by them and by

consumers to do work, and is eventually lost from the ecosystem as heat. Chemicals cycle between organisms and the environment.



Interactions Theme: From Molecules to Ecosystems, Interactions Are Important in Biological Systems

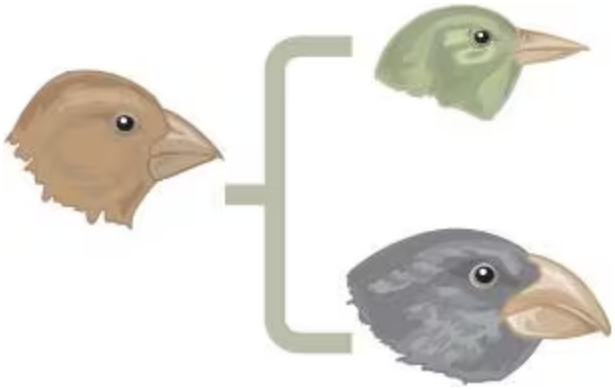
- In **feedback regulation**, a process is regulated by its output or end product. In negative feedback, accumulation of the end product slows its production. In positive feedback, an end product speeds up its own production.
- Organisms interact continuously with physical factors. Plants take up nutrients from the soil and chemicals from the air and use energy from the sun.



Thinking about the muscles and nerves in your hand, how does the activity of text messaging reflect the four unifying themes of biology described in this section?

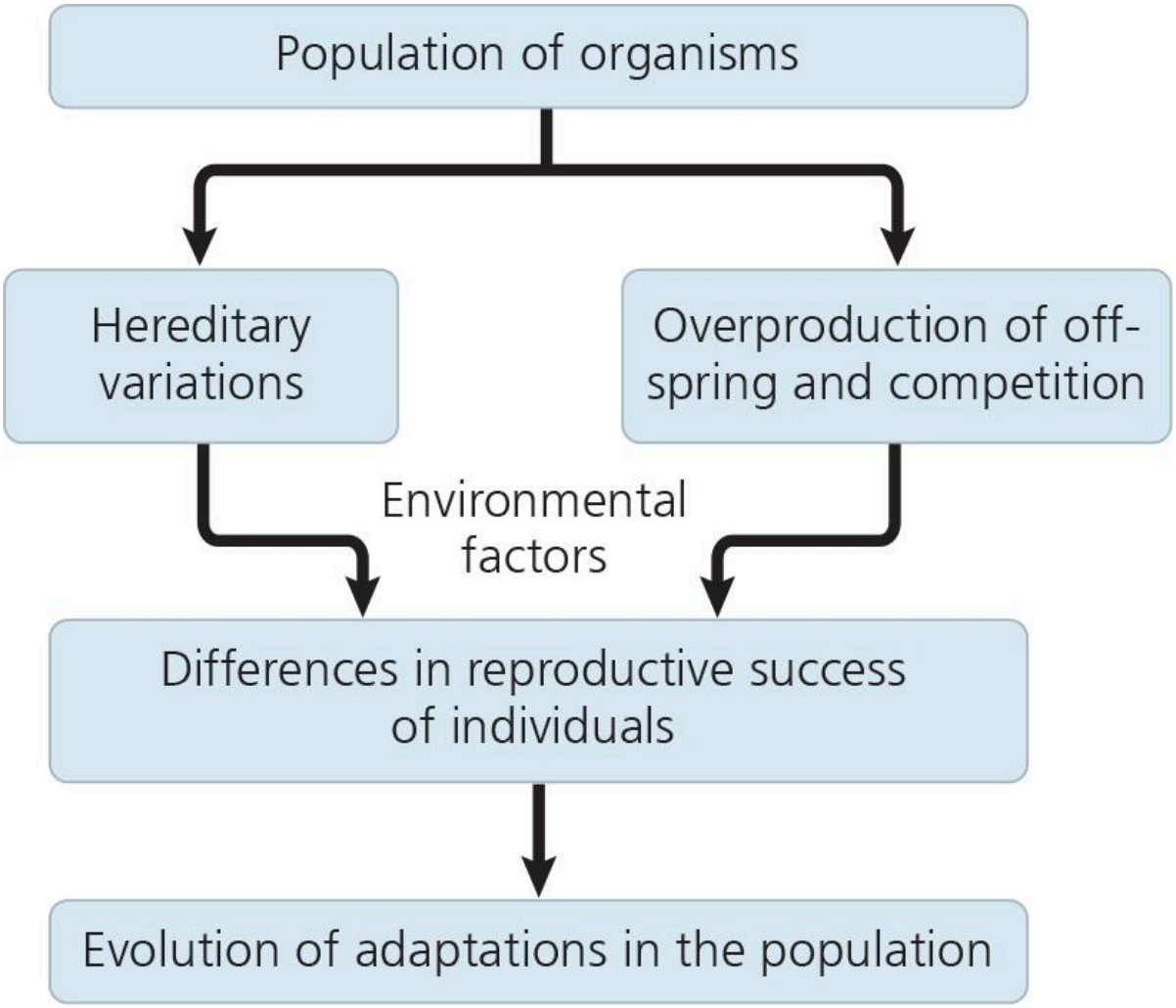
Concept 1.2: The Core Theme: Evolution accounts for the unity and diversity of life

- ****Evolution**, the process of change that has transformed life on Earth, accounts for the unity and diversity of life. It also explains evolutionary adaptation—the match of organisms to their environments.



- Biologists classify species according to a system of broader and broader groups. Domain **Bacteria** and domain **Archaea** consist of prokaryotes. Domain **Eukarya,** the eukaryotes, includes various groups of protists and the kingdoms Plantae, Fungi, and Animalia. As diverse as life is, there is also evidence of remarkable unity, revealed in the similarities between different species.

- Darwin proposed **natural selection** as the mechanism for evolutionary adaptation of populations to their environments. Natural selection is the evolutionary process that occurs when a population is exposed to environmental factors that consistently cause individuals with certain heritable traits to have greater reproductive success than do individuals with other heritable traits.



Within a population of organisms, two factors, hereditary variations and overproduction of offspring and competition, combined with environmental factors, lead to differences in the reproductive success of individuals. Those accumulated differences lead to the evolution of adaptations in the population.

- Each species is one twig of a branching tree of life extending back in time through more and more remote ancestral species. All of life is connected through its long evolutionary history.

How could natural selection have led to the evolution of adaptations such as camouflaging coat color in beach mice?

Concept 1.3: In studying nature, scientists form and test hypotheses

- In scientific **inquiry**, scientists collect **data** and use **inductive reasoning** to draw a general conclusion, which can be developed into a testable **hypothesis**. **Deductive reasoning** uses predictions to test hypotheses. Hypotheses must be testable; science can address neither the possibility of supernatural phenomena nor religious beliefs. Hypotheses can be tested by conducting **experiments** or, when that is not possible, by making observations. In the process of science, the core activity is testing ideas. This endeavor is influenced by exploration and discovery, community analysis and feedback, and societal outcomes.
- **Controlled experiments** are designed to demonstrate the effect of one **variable** by testing control groups and experimental groups that differ in only that one variable.
- A scientific **theory** is broad in scope, generates new hypotheses, and is supported by a large body of evidence.

What are the roles of gathering and interpreting data?

Concept 1.4: Science benefits from a cooperative approach and diverse viewpoints

- Science is a social activity. The work of each scientist builds on the work of others who have come before. Scientists must be able to repeat each other's results, and integrity is key. Biologists approach questions at different levels; their approaches complement each other.
- ****Technology** consists of any method or device that applies scientific knowledge for some specific purpose that affects society. The impact of basic research is not always immediately obvious.
- Diversity among scientists promotes progress in science.

Explain why different approaches and diverse backgrounds among scientists are important.